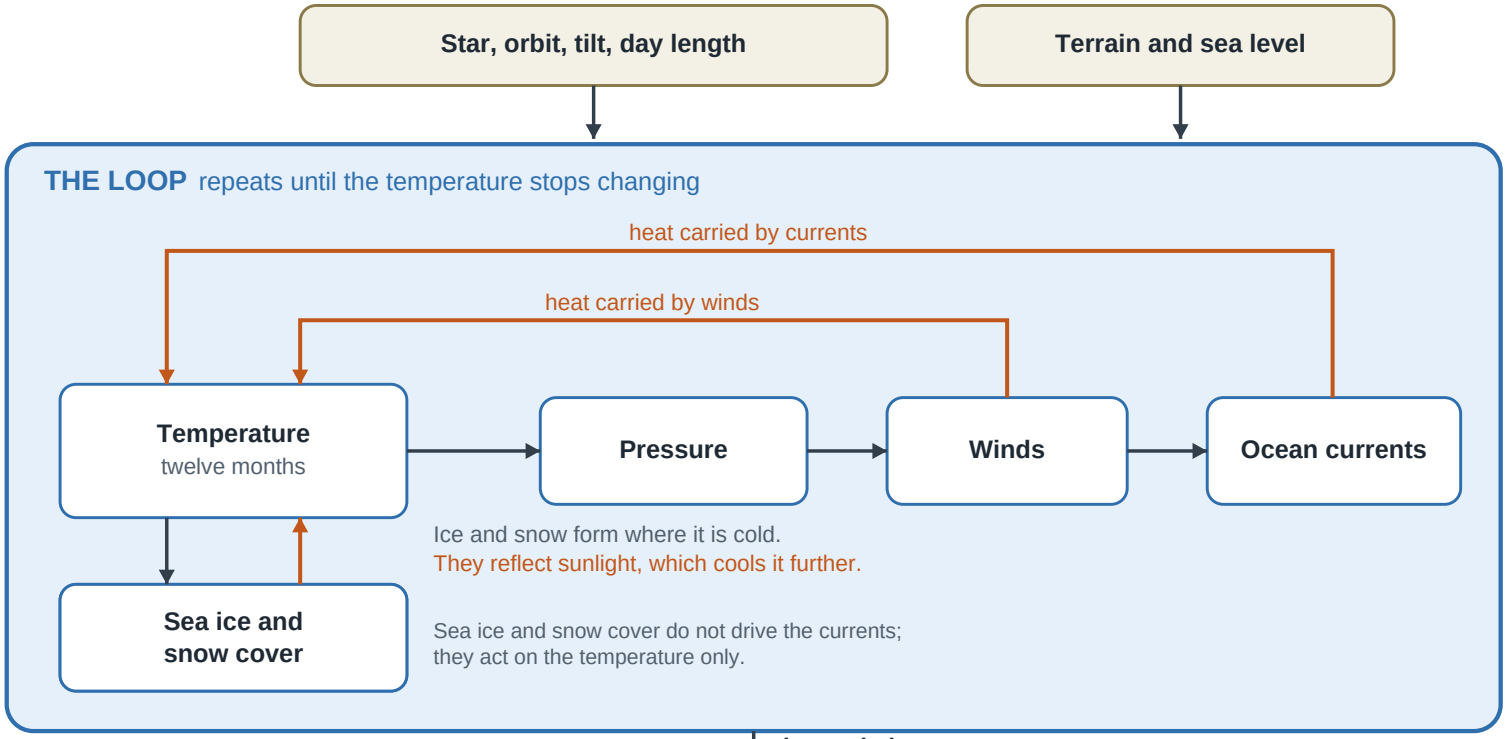
earth_sim: how the climate is worked out

The design once the biome map (roadmap step 8) is built. Drawn 14 November 2026.



Layers you can view

From the loop

- Temperature
- Pressure
- Winds
- Ocean currents
- Sea ice

Downstream

- Rainfall
- Climate zones
- Storm zones
- Biomes

Outside the loop, stepping through time

Ice sheets grow over thousands of years (step 6). Plates move over millions (step 9).

They change the terrain and the ice cover, and the loop is then run again as a snapshot.

→ feeds the next step → feeds back into the temperature

Built today (step 3, in progress): temperature, pressure and winds, worked out once in that order, with no feedback.

Planned, not built: the loop and heat carried by winds (3b), rain (4), ocean currents (5), ice (6), storm zones (7), biomes (8).

The wind model's settings

The 31 numbers the wind model is built on, as of 14 November 2026 (piece 7c-1, in review). In orange: the 11 tuned by hand against the observed Earth.

Setting	Value	What it does	Where it comes from	Tuned?
Width of the cells				
`earthContrastK`	41 K	The Earth's equator-to-pole temperature contrast: the yardstick other planets' contrast is measured against.	The simulated Earth	
`rotationPower`	0.5	How the cells' width follows the day length: the square-root rule.	Published theory (Held, 2000)	
`contrastPower`	0.25	How the cells' width follows the temperature contrast.	A guess from theory	
`minContrastRatio`	0.25	The least contrast taken, so cells keep a sane width on a planet with almost none.	A guess	
`maxContrastRatio`	4	The most contrast taken, so cells keep a sane width on a planet with a huge one.	A guess	
Where the central low sits				
`hotWindowK`	4 K	How the warmest latitude is averaged, so the low does not jump between two warm bands.	Started at 2	**Tuned against the Earth**
`lowShiftGain`	2.5	Pushes the low further from the equator than the warmest latitude. Makes up for the temperature model's weak seasons.	1 is the plain rule	**Tuned against the Earth**
`lowShiftFadeDeg`	20 deg	The latitude over which that push fades away.	A guess	
`edgeShiftShare`	0.5	How far the subtropical highs follow the low off the equator.	The observed Earth	
`lowLimitDeg`	70 deg	The farthest the low may sit from the equator at the Earth's spin or faster.	Our choice, after two papers	
`lowLimitFromWidthDeg`	30 deg	The cell width up to which that limit is 70 degrees: the Earth's.	Our choice	
`lowLimitToWidthDeg`	60 deg	The cell width from which the limit is the pole: a 4-day day.	Our choice, after a paper's slow case	
`lowLimitEaseShare`	0.5	How gently the low is held short of its limit.	A guess	
Steep tilts				
`minSummerCellShare`	0.3	The narrowest the tropical cell may be on the summer side of the low.	A guess	
`wideSummerFromShare`	0.4	How far out the low lies when that summer side starts to widen again.	Set by hand	Set against the 60-degree tilt case
`wideSummerToShare`	0.8	How far out the low lies when the summer side is a whole cell again.	Set by hand	Set against the 60-degree tilt case
`sideContrastK`	30 K	How much cooler a pole must be than the low for that side's bands to be at full strength.	Started at 20	Set against the 60-degree tilt case
Strength of the bands				
`bandAmplitudeHpa`	9 hPa	How strong the pressure bands are, and so how fast the trades and westerlies blow.	Started at 5	**Tuned against the Earth**
`outerCellShare`	0.3	The strength kept by bands far from the low; the real polar easterlies are weak.	Started at 1	**Tuned against the Earth**
`fullBandsReachDeg`	60 deg	On a planet spinning faster than the Earth, how far from the low the bands keep full strength.	Our choice	
`poleTaperDeg`	10 deg	The distance from each pole over which the bands ease out.	Started at 15	Set by hand, not against the Earth

The wind model's settings (continued)

Setting	Value	What it does	Where it comes from	Tuned?
Highs and lows over land and sea				
`thermalHpaPerK`	3 hPa/K	Pressure added for each degree a place is colder than its latitude. Makes up for the temperature model's weak land and sea contrasts.	Physics suggests 0.7	**Tuned against the Earth**
`thermalSmoothingKm`	900 km	How far those highs and lows are smoothed.	Started at 750	**Tuned against the Earth**
`seaThermalShare`	0.3	The share of that response the sea gets, against land.	Was 1	**Tuned against the Earth**
`warmLimitK`	6 K	Caps how deep a hot place's low gets.	No limit before	**Tuned against the Earth**
`coldKneeK`	20 K	A cold place's high grows in full up to this many degrees below its latitude.	A guard for extreme worlds	
`coldLimitK`	25 K	The most cold a high counts, however cold the place.	A guard for extreme worlds	
`thermalHeightM`	2,500 m	Fades the response over high ground, so a cold plateau is not a false high.	No fade before	**Tuned against the Earth**
The wind from the pressure				
`seaFrictionPerS`	2.5e-5 /s	Friction over the sea: how far the wind cuts across the pressure lines.	Started at 3.7e-5	**Tuned against the Earth**
`landFrictionFactor`	5	How many times more friction land has: the wind's speed over land.	Started at 2	**Tuned against the Earth**
`blockingSlope`	0.005	The slope of the ground at which mountains turn half the wind along them.	A guess	

The temperature model's settings

The 18 numbers the temperature model is built on, as of 14 November 2026: 13 physical and 5 for the solver. In orange: tuned against the observed Earth.

Setting	Value	What it does	Where it comes from	Tuned?
How heat moves and is stored				
`diffusivity`	0.55 W/m2/K	How readily heat spreads between neighbouring places.	Published (North, Cahalan and Coakley, 1981), a little below	
`landDiffusivity`	0.35 W/m2/K	The same over land: lower, so continents keep more of their own seasons.	Started at the sea's value	**Tuned against the Earth**
`oceanHeatCapacity`	2.5e8 J/m2/K	The heat the sea stores: a mixed layer of 60 m of sea water.	A physical figure	
`landHeatCapacity`	1e7 J/m2/K	The heat land stores: about the column of air above it alone.	As simple seasonal models take it	
`seaIceHeatCapacity`	1e7 J/m2/K	The heat sea ice stores: as land, since ice insulates the water below.	Chosen; without it the Arctic winter is 10 K too warm	
`lapseRate`	6.5 K/km	How fast it cools with height.	The standard atmosphere	
`greenhouseFactor`	0.633	How much outgoing heat the air holds back at the Earth's greenhouse strength.	The Earth's energy budget	
Reflected sunlight				
`albedo`	0.27	The share of sunlight reflected, on average, by ground clear of ice.	Set to the observed global mean	**Tuned against the Earth**
`albedoPoleward`	0.078	How much more is reflected towards the poles.	Published form, tuned	**Tuned against the Earth**
`landAlbedoExtra`	0	Extra reflection over land. Left at zero, for tuning.	No starting point	
`iceAlbedo`	0.62	The share reflected by ice and snow.	Published form, tuned	**Tuned against the Earth**
`iceStartC`	0 C	The temperature at which ice starts to come in.	A smooth ramp, not a single temperature	
`iceFullC`	-10 C	The temperature at which a place is fully ice.	A smooth ramp, not a single temperature	
The solver (numerical, not physical)				
`stepsPerMonth`	2	Time steps in each month.	Chosen by measurement	
`solverToleranceK`	0.005 K	How closely each step's heat balance is solved.	Chosen by measurement	
`solverMaxIterations`	200	Stops a solve that would not converge; it normally needs 5 to 25.	A safety limit	
`terrainDriftLimitK`	0.25 K	How far the terrain may change before the temperature is solved from nothing again.	Chosen, not derived	
`correctionRefineK`	1 K	The year-end correction below which later passes are skipped.	Chosen by measurement	